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Assignment 1

Literature Review

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1. Introduction

Sentiment Analysis (also known as opinion mining) is a fundamental task within Natural Language Processing which involves detecting, extracting and labelling sentiments expressed by humans in natural language. With the rise of digital communication, microblogging and public online forums, we are now faced with large volumes of text that must be classified using automated methods for brand monitoring, customer feedback, market prediction and political analysis. There are many levels of sentiment classification from simple binary classifications (Positive/Negative) and three-class classifications (Positive/Neutral/Negative) to more complex classifications including multi-class and emotion based (Anger/Fear/Happiness/Love/Sadness/Surprise). Traditional Natural Language Processing techniques show a high level of accuracy when working with well-structured and edited text such as news articles or encyclopaedias. However, applying this to text in social media can present challenges.

Using Sentiment Analysis on social media data presents many issues due to the large amount of noise. The nature of social media means that text will have irregular grammar, misspellings, slang, acronyms and internet slang meaning there are large amounts of Out-of-Vocabulary (OOV) words. For example, in Transliterated Tamil comments there are 2330 OOV words. This causes issues with sub-word tokenizers. Social media often has code-mixing and transliteration meaning that words can be from different languages. Users may write using a mixture of their local dialect and English. There is also a large amount of sentiment expressed through emojis and emoticons. These are often removed during the cleaning process, but they give useful information. It has been proven that emojis carry high amounts of sentiments. In real world social media data there is often an imbalance in the dataset. This causes problems with traditional machine learning models as they will over predict the most common class.

Many researchers have looked at different approaches to tackle these issues. From simple methods involving lexicons to more complex methods such as deep learning and Large Language Models. This review looks at recent papers published between 2020 and 2025 looking at Sentiment Analysis on social media using low resource, cross-lingual and code-mixing methods. This review looks at three different ways of handling the problem. First looking at traditional methods using lexicons to normalise the text and then using machine learning methods such as Support Vector Machines and term frequency-inverse document frequency. Second looking at cross-lingual

methods using Singular Value Decomposition and sequential methods such as BiLSTM and attention. Thirdly, looking at multilingual transformer models using fine tuning (m-BERT, XLM-RoBERTa, IndicSBERT), embedding methods using static LLMs (MiniLM, XLM-R, Universal Sentence Encoder) and handling emojis. This review shows how there is a clear gap showing why we have proposed three different methods.

1.1. Key Methods & Architecture

The methods used in social media sentiment literature can be categorized into using the following four architectures Feature Engineering, Custom Cross-Lingual Alignment Methods, In-domain Pre-trained Language Models, and Contrastive Learning.

1.1.1. Feature Engineering & Normalisation with Lexical Resources

Several papers used a feature engineering approach to handle the informal nature of social media texts, including phonetic spelling and high levels of noise. Advani et al. (2020) propose a feature engineered baseline using Logistic Regression on n-gram features generated using TF-IDF along with sentiment scores from VADER, emoji bias, swearing lexicons and user features. Kusampudi et al. (2021) propose an unsupervised approach to handle the phonetic noise of code-mixed Dravidian languages. This method normalises the data before classification using Logistic Regression, Naive Bayes, SVM and Random Forest (ML) models and MLP (DL). Hande et al. (2021) also perform benchmark comparisons on the KanCMD dataset while Eusha et al. (2024) provide baseline results using Logistic Regression, SVM, XGBoost and majority voting ensemble models.

1.1.2. Cross-Lingual Alignment & Custom Attention Networks

Custom alignment strategies were also proposed to allow transfer learning between languages without requiring full machine translation. Lim and Liew (2022) propose a Hierarchical Multi-Head Attention Network (HMAN) which utilises SVD to align the word vectors of English and Malay. This method also includes using Extended Affective Norms for English Words (E-ANEW) valence scores to enhance the alignment of the words with similar sentiment. To handle code-switching, Romadhona et al. (2022) propose Mixed XLM which includes language specific

features. Token level language features are used along with code-mixed phrase level data augmentation. Sharma et al. (2023) propose a late fusion approach where the output of multiple fine-tuned models (BERT, m-BERT, XLM-R) are placed into an MLP. This allows the model to learn weights to combine the outputs.

1.1.3. In-domain Continual Pre-training & Pre-trained LLM Encoders

To handle the slang within social media texts, continual pre-training was performed on microblog data. A continual pre-training approach was proposed with XLM-T (Barbieri et al., 2022), which extends on XLM-RoBERTa. This model is continually pre-trained using masked language modelling (MLM) on 198 million multilingual tweets. Camacho-Collados et al. (2022) propose TweetNLP which includes a RoBERTa and XLM-RoBERTa model which were pre-trained on Twitter data (TimeLMs-21 as the base English model). Dadure et al. (2025) investigate a two stage feature extraction approach. Dense sentence embeddings are extracted using different pre-trained LLMs (MiniLM-L12-v2, XLM-RoBERTa base, Universal Sentence Encoder). These embeddings are then used to train ML models (Logistic Regression, SVM, Random Forest, Decision Tree, KNN).

1.1.4. Contrastive Learning & Transformer Ensemble

To ensure models can generalise well on low resource and imbalanced sentiment data, state of the art models use fine-tuned multilingual transformer encoders. Hasan et al. (2023) investigated the best performing models for the BLP-2023 sentiment challenge for Bangla. There was a large improvement in performance when fine-tuned models (BanglaBERT, m-BERT, XLM-RoBERTa) were combined into an ensemble along with adversarial weight perturbation (AWP) and pseudo-labelling. Hu et al. (2023) propose Sentiment-Aware Contrastive Learning (SACL) using XLM-RoBERTa. Adversarial noise is added to the model along with supervised contrastive loss to bring similar sentiment states together.

1.2. Datasets & Languages

The data used within the literature include monolingual low resource corpora, regional code-mixed and transliterated data, and large-scale multilingual microblogging data. Monolingual low resource corpora have been built to capture regional sentiment nuances. Lim and Liew (2022) used a portion of single-labelled tweets (10,904) from EmoTweet-28 as the source data for cross lingual emotion transfer. The target data was a gold standard dataset of 1,843 tweets in Malaysian Malay taken from the Malaya Twitter dataset, which were labelled into seven fine grained emotion classes. Dadure et al. (2025) used the Arabic Sentiment Twitter Corpus consisting of 56,795 tweets, collected in March and April 2019. This was a balanced dataset of 28,513 positive and 28,282 negative tweets which contained sentiment dialects and a large number of emoticons. Hasan et al. (2023) merged MUBASE (Facebook and Twitter news comments) and SentNoB (multi-topic comments from newspapers and YouTube in 13 domains) into a Bangla sentiment dataset containing 49,333 samples of Bangla with 3-class sentiment. Informal code mixing and transliteration data has been widely studied to overcome the issue of people mixing their languages in online interactions. Eusha et al. (2024) studied 4-class sentiment in code mixed comments on YouTube. They used 20,070/733 training/test samples of Tamil-English code mix and 3,352/531 training/test samples of Tulu-English. However, there was a large class imbalance with over two thirds of the training data being positive. The testing datasets had a large out of vocabulary rate. Kusampudi et al. (2021) developed the CMTET corpus which is a code mixed dataset of 19,857 Telugu-English sentences. Hande et al. (2021) developed the KanCMD benchmark for sentiment analysis of code-mixed Kannada-English comments on YouTube consisting of 7,671 samples. Advani et al. (2020) used the SemEval-2020 SentiMix benchmark to study sentiment analysis across code mixing in Indic and Hispanic languages. This included Hinglish (20,000 Hindi-English tweets) and Spanglish (18,788 Spanish-English tweets). Sharma et al. (2023) used the GLUECoS benchmark which contains 10,080 English-Hindi tweets and 1,523 English-Spanish tweets. Romadhona et al. (2022) developed the BBRC pre-training dataset by creating code mix data through phrase swapping. They also developed the SentiBahasaRojak dataset which contains human verified sentiment information from code mixed Malay, English and Chinese tweets relating to movies and stocks. Large scale multilingual microblogging data has been used throughout multiple studies to test multi-platform performance. Barbieri et al. (2022) and Camacho-Collados et al. (2022) used the TweetEval benchmark for English microblogging data,

and the UMSAB benchmark which contains tweets in 24,262 tweets in 8 different languages (Arabic, English, French, German, Hindi, Italian, Portuguese, Spanish). There were also pre-training corpora for over 30 languages. Hu et al. (2023) used the AfriSenti-SemEval benchmark to study cross linguistic transfer in low resource African languages. This contained over 110,000 tweets in 14 low resource languages (Hausa, Yoruba, Amharic etc.) and zero shot tracking in languages like Tigrinya and Oromo.

1.3. Sentiment Granularity

The literature reveals the use of both basic binary classification for the sentiment labels to more complex multi-class and fine-grained emotion classification. Most works use a 3-class coarse-grained sentiment classification system (Positive, Negative, Neutral), including Hasan et al. (2023), Camacho-Collados et al. (2022), Sharma et al. (2023), Advani et al. (2020), Barbieri et al. (2022), Kusampudi et al. (2021), and Hu et al. (2023). However, some researchers use basic binary classification (Positive, Negative) for their research (Dadure et al., 2025) and (Romadhona et al., 2022). For more accurate representation of opinions in the real world, multi-class systems have been used within recent works (4 classes Positive, Negative, Neutral, Mixed) (Eusha et al., 2024) and (5 classes Positive, Negative, Neutral, Mixed, Not-Kannada) (Hande et al., 2021). Furthermore, fine-grained emotion classification has been conducted with 7 classes (anger, fear, happiness, love, none, sadness, surprise) (Lim and Liew, 2022).

1.4. Preprocessing & Non-Textual Features

As social media data can be quite noisy, a wide range of preprocessing methods have been used in the literature. Researchers have used multiple text normalisation methods to improve the quality of the text in low resource and code-mixed languages (Lim & Liew, 2022), (Romadhona et al., 2022). This includes normalising contractions, looking up regional Malay slang and expanding abbreviations (Lim & Liew, 2022). Others have developed unsupervised methods for code-mixed languages (Kusampudi et al., 2021). They created a way to normalise the data using clustering based on Levenshtein distance and merging aspirated consonants. Some have normalised transliterated text for the languages Hindi and Kannada (Sharma et al., 2023) and (Hande et al., 2021). Other researchers have expanded abbreviations and merged any repeated characters in the data (Advani et al., 2020). Most works have also carried out a preprocessing step to clean the data in some way (Hasan et al., 2023), (Camacho-Collados et al., 2022) and (Barbieri et al., 2022). A key difference was that some removed non-textual information, like emojis, punctuation and special characters, as it can be random and not relevant to anything (Eusha et al., 2024), (Hande et al., 2021). However, others have chosen to keep emojis in their data (Dadure et al., 2025), (Barbieri et al., 2022). Some works have retained the polarity provided by emojis (Advani et al., 2020). Others have included a method to convert emojis to text (Lim & Liew, 2022). Some have even created their own benchmark based on predicting emojis (Camacho-Collados et al., 2022). Researchers have also added sentiment prefixes to the text for their specific language (Hu et al., 2023). They showed that text normalisation along with retaining non-textual features can be very effective at overcoming the ambiguity in short social media texts.

1.5. Key Findings & Performance

The results from the literature show social media sentiment analysis can greatly benefit from domain pre-training and fine-tuning. For cross-lingual emotion analysis, Lim & Liew (2022) achieved a Macro F1 score of 0.8836 with fine-tuning their Hierarchical Multi-Head Attention Network (HMAN) in few shots. This compares well with mBERT (0.8925) with their model running 6-7 times faster. The literature also shows varying results for different target languages in low resource and code-mixed languages. Eusha et al. (2024) showed m-BERT achieved the best result (1 st place) with a Macro F1 score of 0.58 for Tulu. However, the imbalance in their data resulted in poor results for Tamil with the highest result being only 0.23 Macro F1 (L3Cube-IndicSBERT and classical ML tied). Overall, retaining non-textual emotional features greatly improved results. Dadure et al. (2025) achieved a high accuracy of 85.98% using their system with MiniLM-L12-v2 sentence embeddings and Random Forest classifier. This compared to 78.52% without emojis.

Overall, fine-tuned transformer ensembles and domain pre-training achieved good results. Hasan et al. (2023)'s winning system Aambela achieved 0.7310 micro-F1 on Bangla social media. The domain adapted XLM-T achieved good results across multiple languages with a Macro F1 score of 67.91%-69.35% on the multilingual UMSAB benchmark (Barbieri et al., 2022) and (Camacho-Collados et al., 2022). This was better than the generic XLM-R. Some works have shown late fusion methods with neural networks achieved good results for code-mixed languages. Sharma et al. (2023) achieved state-of-the-art results with F1 scores of 73.66% for English-Spanish and 61.24% for English-Hindi. Romadhona et al. (2022) have also shown Mixed XLM achieved good results with an F1 score of 0.705 for Bahasa Rojak. This was better than XLM (0.637). Text normalisation and regularisation methods have also helped improve results. Kusampudi et al. (2021) achieved great improvements with up to 2.53% increase in accuracy using their unsupervised phonetic normalisation. Advani et al. (2020) achieved good results with their feature engineered Logistic Regression. They achieved a weighed F1 score of 0.65 for Hinglish and 0.63 for Spanglish. Hande et al. (2021) showed mBERT outperformed classical models for Kannada-English comments. Hu et al. (2023) achieved a first-place result for the AfriSenti zero-shot subtask. They used SACL-XLMR to achieve 1.1% improvement in multilingual and 2.8% improvement in zero-shot classification.

2. Thematic Discussion of Past Studies

To evaluate existing research without listing publications in isolation, past studies are grouped logically into three primary themes reflecting the methodological evolution of social media sentiment analysis.

2.1.Theme 1: Traditional Lexicon-Assisted Normalization & Feature-Engineered Baselines

Early computational approaches to social media sentiment and emotion analysis relied heavily on rule-based text cleaning, affect dictionaries, and statistical feature engineering paired with classical machine learning algorithms. Because informal social media text contains spontaneous syntax, single-letter negations, and dialectal shorthand, baseline pipelines implement systematic normalization expanding contractions, converting emoticons to text, and filtering non-standard vocabulary against formal standard dictionaries. To represent text numerically, classical systems extract boolean n-grams or term frequency-inverse document frequency (TF-IDF) matrices and pass them to linear classifiers such as Support Vector Machines (SVM), Logistic Regression with class-weighted penalties, Random Forests, or majority voting ensembles. Additionally, lexicon-assisted frameworks incorporate pre-compiled affect dictionaries (e.g. General Inquirer or LIWC) to substitute explicit emotion words with generic sentiment tags, or use bilingual dictionaries as anchor matrices for Singular Value Decomposition (SVD) vector rotation and valence score refinement. Although feature-engineered baselines offer low computational overhead and fast CPU execution, they suffer from fundamental failure modes: they rely on local n-gram co-occurrence without modeling long-range contextual dependencies, fail on unlisted social media slang, and struggle to resolve complex negation or sarcasm.

2.2.Theme 2: Cross-Lingual Vector Space Alignment & Sequential Deep Learning

Cross-lingual vector space alignment and sequential deep learning have improved social media sentiment analysis, particularly for multilingual and low-resource languages. Lim and Liew (2022) aligned English-Malay word embeddings using Singular Value Decomposition (SVD) and incorporated them into a Hierarchical Multi-head Attention

Network (HMAN). Their findings demonstrated that emotion knowledge could be transferred from English to Malay with limited Malay training data, reducing dependence on machine translation. Similarly, Kusampudi et al. (2021) applied unsupervised normalization to Telugu-English text to address spelling variations and transliteration inconsistencies, showing that improved input quality could enhance classification performance. Eusha et al. (2024) compared machine learning, BiLSTM, CNN, and Transformer models for Tamil-English and Tulu-English text. Their results showed that the best-performing models differed between languages, highlighting the influence of language characteristics and available training data. Romadhona et al. (2022) developed Mixed XLM and the SentiBahasaRojak dataset to address Malay-English-Chinese code-mixing, while Sharma et al. (2023) combined the outputs of two Transformer models for English-Hindi and English-Spanish sentiment analysis. Collectively, these studies demonstrate the value of cross-lingual representations, text normalization, and model fusion for handling multilingual text. However, performance remains affected by limited language resources, spelling variations, class imbalance, and differences between languages. These challenges motivate further research into pretrained contextual models that can capture richer linguistic information.

2.3.Theme 3: Pre-Trained Contextual Transformers, LLM Encoders, & Non-Textual Feature Retention

Pretrained contextual Transformers have advanced sentiment analysis by learning contextual representations from large text collections rather than relying solely on manually engineered features. Camacho-Collados et al. (2022) introduced TweetNLP, a toolkit providing models adapted to social media text, while Barbieri et al. (2022) developed XLM-T by further pretraining XLM-R on multilingual tweets. These studies demonstrate the importance of domain-specific pretraining for handling informal expressions, hashtags, abbreviations, and other characteristics of social media. Hasan et al. (2023) explored traditional machine learning, fine-tuned language models, and zero-shot or few-shot approaches for Bangla sentiment analysis, with the best-ranked system achieving a micro-F1 score of 0.7310. Hu et al. (2023) used sentiment lexicon information

and contrastive learning to improve multilingual and zero-shot sentiment classification for African languages, achieving first place in the zero-shot subtask. Eusha et al. (2024) also found that the best model varied between Tamil and Tulu, indicating that multilingual models do not perform equally well across all languages. In addition to model selection, retaining useful sentiment cues is important. Dadure et al. (2025) investigated Arabic tweet classification using pretrained sentence encoders and reported different results across configurations involving different encoders and emoji treatment. Advani et al. (2020) likewise incorporated emoji polarity and punctuation into handcrafted features. These studies suggest that both contextual representations and sentiment-bearing features can contribute to classification. Nevertheless, pretrained models continue to face challenges involving low-resource languages, class imbalance, sarcasm, computational cost, and generalisation across datasets. Therefore, model performance should be evaluated alongside preprocessing choices and the characteristics of the target language and dataset.

3. Comparative analysis of approaches & architectural trade-offs

A rigorous comparison across the three paradigms highlights critical trade-offs between computational complexity, context representation, preprocessing strategy, and model adaptability:

Comparative Dimension	**Traditional Lexicon / ML (Theme 1)**	**Vector Alignment & Deep Learning (Theme 2)**	**Transformer & LLM Encoders (Theme 3)**
Representative Models	Logistic Regression, SVM, Naive Bayes, Random Forest, Decision Tree, KNN, and XGBoost paired with VADER or phonetic normalization (Advani et al., 2020; Kusampudi et al., 2021; Eusha et al., 2024).	HMAN with SVD-aligned embeddings and E-ANEW (Lim & Liew, 2022); CNN, BiLSTM + fastText/Word2Vec (Hande et al., 2021; Eusha et al., 2024); MLP with phonetic normalization (Kusampudi et al., 2021).	Fine-tuned PLMs: mBERT, XLM-R, XLM-T, RoBERTa/TimeLMs-21, BanglaBERT, MuRIL, IndicSBERT, Mixed XLM, SACL-XLMR, Late-Fusion MLP, and frozen MiniLM-L12-v2/USE encoders (Barbieri et al., 2022; Camacho-Collados et al., 2022; Romadhona et al.,

			2022; Hasan et al., 2023; Hu et al., 2023; Sharma et al., 2023; Dadure et al., 2025).
Feature Representation	Sparse, high-dimensional surface representations using word or character TF-IDF n-grams, boolean bag-of-words, handcrafted sentiment lexicons, profanity counts, and metadata features (Advani et al., 2020).	Static or cross-lingually rotated dense word vectors (Word2Vec, fastText, SVD-aligned embeddings refined with affective valence norms) processed sequentially or via hierarchical attention (Lim & Liew, 2022; Eusha et al., 2024).	Dynamic, bidirectional contextualized subword representations via self-attention, token-level language embeddings (Romadhona et al., 2022), contrastive sentiment clusters (Hu et al., 2023), or dense sentence-level LLM embeddings (Dadure et al., 2025).
Preprocessing & Emojis	Heavy manual normalization required: abbreviation expansion, Levenshtein phonetic clustering, stop-word removal, and explicit extraction of emoji polarity scores or emoticon-to-text mapping (Advani et al., 2020; Kusampudi et al., 2021).	Moderate normalization: contraction expansion, Malay slang spell-checking, Roman script transliteration, and converting emojis to text or emoticons or stripping them (Hande et al., 2021; Lim & Liew, 2022; Eusha et al., 2024).	Minimal structural cleaning (removing URLs/HTML tags) while preserving native social media syntax, slang, and emojis directly as semantic subword tokens (Barbieri et al., 2022; Camacho-Collados et al., 2022; Dadure et al., 2025), alongside lexicon prefix injection (Hu et al., 2023).
Computational Cost	Low cost because fast training and inference executable on standard CPU hardware with minimal memory footprint and no GPU pre-training dependency (Advani et al., 2020).	Moderate cost because equires GPU acceleration for sequential or attention training, but avoids massive parameter overhead. Example, HMAN executes 6–7× faster than mBERT (Lim & Liew, 2022).	High cost because Continual pre-training on 198M tweets (Barbieri et al., 2022) and multi-model ensembles or late fusion (Hasan et al., 2023; Sharma et al., 2023) demand substantial GPU

			memory and training time, though frozen LLM feature extraction (Dadure et al., 2025) reduces fine-tuning cost.
Context & OOV Handling	Captures only local n-gram co-occurrence completely blind to long-range syntax, though character n-grams and phonetic clustering partially mitigate spelling variants (Advani et al., 2020; Kusampudi et al., 2021).	Captures sequential and hierarchical sentence relationships, but relies on static bilingual dictionaries and suffers when word layers are frozen or encounter unseen transliterations (Lim & Liew, 2022; Eusha et al., 2024).	Self-attention resolves long-range dependencies and polysemy; Byte-Pair Encoding (BPE) and WordPiece handle rare words via subwords, but severe unnormalized transliteration (e.g., 2,330 OOV words in Tamil) still degrades subword semantics (Eusha et al., 2024).
Primary Failure Mode	Severe feature engineering bottleneck like inability to generalize across languages without rebuilding lexicons; fails on implicit sentiment, complex negation, and sarcasm (Advani et al., 2020).	Dependence on bilingual seed lexicons for vector space alignment (Lim & Liew, 2022) and vulnerability to class imbalance and unnormalized phonetic variations in code-mixed scripts (Hande et al., 2021; Eusha et al., 2024).	Overfitting on majority classes under extreme dataset skewness (Eusha et al., 2024), representation dilution across typologically diverse languages (Barbieri et al., 2022), and persistent difficulty interpreting thread-dependent sarcasm or mixed emotions (Camacho-Collados et al., 2022; Hasan et al., 2023).

4. Systemic limitations of existing work

Despite the domain of social media sentiment analysis having shown a considerable amount of progress, the reviewed studies have also revealed several limitations that remain unsolved across different modelling approaches. Traditional machine learning methods, cross-lingual models, and pretrained Transformers have each addressed specific challenges, but their performances are heavily influenced by language resources, dataset characteristics, preprocessing decisions, and the availability of sentiment-related information.

One major limitation is the difficulty of generalizing across low-resource, multilingual, and code-mixed languages. Although multilingual and cross-lingual models reduce the dependence on large language-specific datasets, their performances are still easily influenced by limited training data, transliteration, spelling variation, and out-of-vocabulary expressions. For example, Eusha et al. (2024) reported substantially different performance across Tamil-English and Tulu-English despite evaluating similar multilingual architectures. Studies by Lim and Liew (2022), Romadhona et al. (2022), and Hu et al. (2023) have also indicated that cross-lingual transfer can improve performance, but it does not completely remove the dependence on language-specific resources. Therefore, multilingual modelling alone is unable to ensure consistent performance across all low-resource languages.

The second limitation is the imbalance of classes in social media sentiment datasets. Some datasets contain substantially more examples from one sentiment class, particularly the positive class, rather than balance between negative, neutral or mixed classes. This can cause classifiers to favor and bias toward majority class and reduce their ability to correctly identify minority sentiments. For example, Eusha at al. (2024) reported severe class imbalance in their code-mixed datasets, while Hande et al. (2021) identified a strong positive-class bias in the KanCMD dataset. Following these conditions, overall accuracy might give an overly optimistic view of model performance because good predictions on majority class can hide weak performance on minority classes. Hence, metrics such as Macro-F1 and per-class-precision and recall are important for evaluating imbalanced sentiment datasets.

Another important limitation is the inconsistency treatment in non-textual sentiment information during preprocessing. Social media posts frequently contain emojis, emoticons, punctuation, repeated characters, and other informal features that may express emotional meaning. However, the reviewed studies do not treat these features consistently. Eusha et al. (2024) and Hande et al. (2021) removed several non-textual elements during preprocessing, while Advani et al. (2020) retained emoji polarity and punctuation as one of the features. Lim and Liew (2022) converted emojis into textual representations, and Dadure et al. (2025) explicitly retained emojis as semantic information. Dadure et al. (2025) reported an accuracy of 85.98% when emojis were retained completely while the accuracy drops to 78.52% when they are removed. This suggests that aggressive preprocessing may unintentionally discard or remove useful sentiment signals. Therefore, treating all non-textual information as noise may reduce the information available to the classifier.

Other than that, contextual ambiguity and limited domain generalization remain a challenge. Although Transformer-based models such as mBERT, XLM-R and XLM-T are able to provide stronger contextual representations than traditional feature-engineered approaches, but remain difficult to interpret sarcasm, irony, slang, and highly informal expressions consistently. Furthermore, many studies evaluate models using a specific platform, language, or benchmark. A model that performs well on Twitter data may not necessarily achieve the same performance on YouTube comments, product reviews, or other social media domains. This implies that high benchmark performance does not automatically imply strong real-word generalization.

Overall, the existing literature shows that no single approach consistently resolves all challenges associated with social media sentiment analysis. Low-resource languages, noisy text, class imbalance, contextual ambiguity and inconsistent preprocessing remain important limitations. Among these issues, class imbalance and the treatment of sentiment-bearing non-textual features are particularly suitable for further analysis and investigation, as these both can directly affect the information available to the classifier and the reliability of its predictions. Among these limitations,

class imbalance and the treatment of sentiment-bearing non-textual features are worth for further investigation as both repeatedly appeared across existing studies and directly influence the reliability of sentiment classification.

5. Proposed research direction and research gap

The reviewed studies demonstrate that both preprocessing choices and dataset characteristics can have a significant amount of influence on sentiment classification performance. However, within the reviewed studies, the effects of non-textual sentiment retention and class imbalance handling are generally evaluated under different datasets, models, preprocessing processes, and experimental conditions. This makes it difficult to determine the independent contributions of each factor towards classification performance. Hence, the literature reveals a practical research gap in a controlled evaluation of these two factors under the same experiment settings.

To address the gap, the proposed study will focus on two recurring issues identified in the reviewed studies, the treatment of emoji-based sentiment information, and the bias introduced by imbalanced sentiment classes. The study will investigate on whether preserving sentiment-bearing emoji information and addressing class imbalance can improve sentiment classification performance under a consistent and lightweight experimental setting. This direction allows the effects of these factors to be examined more clearly without introducing unncessary architectual complexity.

The proposed study does not aim to introduce a new NLP architecture. Instead, the main focus is to provide a controlled and reproducible comparison of two practical factors that are treated inconsistently in the existing literature, which are sentiment-bearing non-textual information and class imbalance. The study is expected to clarify whether preserving emoji information, applying balanced class weighting, or combining both approaches produces more reliable sentiment classification on noisy social media text. This focused design also keeps the implementation feasible while directly addressing the limitations identified in the reviewed studies.

References

1. Advani, L., Sharma, L., & Mamidi, R. (2020). C1 at SemEval-2020 Task 9: SentiMix: Sentiment analysis for code-mixed social media text using feature engineering. In *Proceedings of the Fourteenth Workshop on Semantic Evaluation*, 1084–1090. https://aclanthology.org/2020.semeval-1.163.pdf

2. Barbieri, F., Camacho-Collados, J., Espinosa-Anke, L., & Neves, L. (2022). XLM-T: Multilingual language models in Twitter for sentiment analysis and beyond. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, 2582–2594. https://aclanthology.org/2022.lrec-1.27.pdf

3. Camacho-Collados, J., Rezaee, K., Riahi, T., Ushio, A., Loureiro, D., Antypas, D., Boisson, J., Espinosa-Anke, L., Liu, F., Martínez-Cámara, E., Medina, G., Buhrmann, T., Neves, L., & Barbieri, F. (2022). TweetNLP: Cutting-edge natural language processing for social media. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 38–49. https://aclanthology.org/2022.emnlp-demos.5.pdf

4. Dadure, P., Dixit, A., Tewatia, K., Paliwal, N., & Malla, A. (2025). Sentiment analysis of Arabic tweets using Large Language Models. In *Proceedings of the 1st Workshop on Arabic Natural Language Processing (AbjadNLP)*, 88–94. https://aclanthology.org/2025.abjadnlp-1.10.pdf

5. Eusha, A. H., Farsi, S., Hossain, A. I. J., Ahsan, S., & Hoque, M. M. (2024). CUET_Binary_Hackers@DravidianLangTech-EACL 2024: Sentiment analysis using transformer-based models in code-mixed and transliterated Tamil and Tulu. In *Proceedings of the Fourth Workshop on Speech, Vision, and Language Technologies for Dravidian Languages*, 205–211. https://aclanthology.org/2024.dravidianlangtech-1.34.pdf

6. Hande, A., Priyadharshini, R., & Chakravarthi, B. R. (2021). KanCMD: Kannada code-mixed dataset for sentiment analysis and offensive language detection. In *Proceedings of the International Conference on Recent Advances in Natural Language Processing (RANLP 2021)*, 450–457. https://aclanthology.org/2021.ranlp-1.86.pdf

7. Hasan, M. A., Alam, F., Anjum, A., Das, S., & Anjum, A. (2023). BLP-2023 Task 2: Sentiment analysis. In *Proceedings of the 1st International Workshop on Bangla Language Processing (BLP-2023)*, 354–363. https://aclanthology.org/2023.banglalp-1.48.pdf

8. Hu, Y., Zhang, Y., & Wang, X. (2023). UCAS-IIE-NLP at SemEval-2023 Task 12: Enhancing generalization of multilingual BERT for low-resource sentiment analysis. In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, 1851–1856. https://aclanthology.org/2023.semeval-1.255.pdf

9. Kusampudi, S. A., Sitaram, S., & Sharma, D. M. (2021). Sentiment analysis in code-mixed Telugu-English text with unsupervised data normalization. In *Proceedings of the First Workshop on Speech and Language Technologies for Dravidian Languages*, 44–50. https://aclanthology.org/2021.dravidianlangtech-1.7.pdf

10. Lim, Y. H., & Liew, J. S. Y. (2022). English-Malay word embeddings alignment for cross-lingual emotion classification with hierarchical attention network. In *Proceedings of the 12th Workshop on Computational Approaches to Subjectivity, Sentiment & Social Media Analysis (WASSA)*, 113–121. https://aclanthology.org/2022.wassa-1.12.pdf

11. Romadhona, N. F., Purwarianti, A., & Lestari, D. P. (2022). BRCC and SentiBahasaRojak: The first Bahasa Rojak corpus for pretraining and sentiment analysis dataset. In *Proceedings of the 29th International Conference on Computational Linguistics (COLING)*, 4426–4434. https://aclanthology.org/2022.coling-1.389.pdf

12. Sharma, G., Chinmay, R., & Sharma, R. (2023). Late fusion of transformers for sentiment analysis of code-switched data. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 6486–6490. https://aclanthology.org/2023.findings-emnlp.430.pdf